B. Evanescent

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: implementation, strings, two pointers | https://codeforces.com/problemset/problem/2254/B

Let $f(s)$ be the compressed version of a string s, formed by replacing every maximal contiguous block of identical characters with a single copy of that character. For example, $f(\texttt{"aabbcc"}) = \texttt{"abc"}$.

Let $|s|$ denote the length of a string s. Following this, $|f(s)|$ denotes the length of the compressed string. For example:

- $|f(\texttt{"aabbcc"})| = |\texttt{"abc"}| = 3$
- If the string is empty, its length is 0.

Yousef has given you a string s consisting of n lowercase Latin letters. You must delete **exactly** one character s_i ($2 \le i \le n - 1$) to form a new string s', and then find the minimum possible value of $|f(s')|$.

Note that you **cannot** delete s_1 or s_n.

Input

The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains an integer n ($3 \le n \le 2 \cdot 10^5$) — the length of the string.

The second line of each test case contains a string s ($|s| = n$), consisting of lowercase Latin letters.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the minimum possible length of the resulting compressed string after deleting one character.

Example

Input

```
9
3
abb
3
aab
3
abc
4
abaa
4
abba
5
eeeee
6
yyssee
7
abacaba
18
goodluckandhavefun
```

Output

```
2
2
2
1
3
1
3
5
16
```

Note

In the first test case, we can only delete the character s_2 = 'b', producing a string s' = "ab", with $|f(s')| = 2$. Therefore, 2 is the minimum length achievable.

In the fourth test case, we can delete the character s_2 = 'b'. The resulting string is s' = "aaa" with $f(s')$ = "a", so $|f(s')| = 1$.

In the sixth test case, deleting any valid character results in $f(s')$ = "e" and $|f(s')| = 1$.

In the eighth test case, we can delete the character s_4 = 'c'. The resulting string is s' = "abaaba" with $f(s')$ = "ababa" and $|f(s')| = 5$.